# Exact Normalized Retained Kernels
# Target 54 of the H1 Downstream Programme

Yang–Mills representational dynamics research programme

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**Abstract**

We construct the exact finite conditional-disintegration object required by the vacuum-weight programme. Normalization, bad-row formation, weighted partition-ratio identities, and compatible relabeling covariance are proved without an independence assumption. The four-channel H1 star kernel is realized exactly by a normalized table. A counterexample shows that an absolutely small bad numerator is useless when the retained-state normalizer collapses. The complete nonlinear boundary-uniform $SU(2)$ kernel remains open.

## 1 Conditional object

Let $S$ be a finite retained-state space and $F_s$ a finite conditional fiber. For $w_s(f) > 0$, a next-state map $\tau_s : F_s \to S$, and bad predicate $b_s : F_s \to \{0, 1\}$, set

$$Z(s) = \sum_{f \in F_s} w_s(f), \quad P(s,t) = \frac{1}{Z(s)} \sum_{\tau_s(f)=t} w_s(f), \quad Q(s,t) = \frac{1}{Z(s)} \sum_{\substack{\tau_s(f)=t \\ b_s(f)=1}} w_s(f).$$

**Theorem 1** (Normalized pushforward)**.** *$P$ is stochastic and $Q$ is substochastic. For every $V : S \to [1, \infty)$,*

$$\frac{(QV)(s)}{V(s)} = \frac{\sum_{b_s(f)=1} w_s(f) V(\tau_s(f))}{Z(s)V(s)}. \tag{1}$$

*Proof.* Sum first over target states and then over fibers. Positivity gives $0 \le Q \le P$, while every fiber occurs exactly once in the sum defining $P$. □

Compatible bijections of states and fibers transport $w, \tau, b$ and conjugate $P, Q$. This is a finite relabeling theorem; physical gauge covariance still requires the boundary algebra and Haar disintegration.

## 2 Exact four-channel realization

Let $S = \{0, 1, 2, 3, 4\}$, with 0 regular, and set $e = q^2/4$. At state 0 use one good fiber of weight $1 - q^2$ returning to 0 and four bad fibers of weight $e$ entering states $1, \dots, 4$. At each charged state use one bad fiber of unit weight returning to 0. Then

$$Q_{0a} = \frac{q^2}{4}, \qquad Q_{a0} = 1, qquadV = (1, q^{-1}, q^{-1}, q^{-1}, q^{-1}),$$

and equation (1) gives $(QV)/V = q$ in every state. For

$$q = 0.002430277501841998$$

the executable identity error is exactly zero in binary floating arithmetic.

## 3 Normalizer obstruction

**Proposition 1.** *Absolute suppression of a bad numerator does not imply conditional suppression.*

*Proof.* Take $Z_G(L) = e^{-2L}$ and $Z_B(L) = e^{-L}$. Both vanish, but

$$\frac{Z_B(L)}{Z_G(L) + Z_B(L)} = \frac{1}{1 + e^{-L}} \longrightarrow 1.$$

At $L = 12$, the bad weight is $6.1442 \times 10^{-6}$ while its normalized probability is 0.9999938558. □

**Warning 1.** *No action-only estimate may replace the retained-state partition normalizer. The physical target is a compact nonlinear $SU(2)$ fiber, uniformly over positive-mass histories, volume, cutoff, charts, center sectors, orbit/cut data, and integer topology.*

## 4 Audit

The exact finite theorem is C2. The physical nonlinear kernel, its uniform normalizer, one-scale basin entry, repeated RG, continuum construction, and a mass gap are not proved. Nine hostile regression tests verify normalization, weighted identities, covariance, invalid-input rejection, and the collapsing normalizer counterexample.